

Mini Project Report
On
CO2 Emission Analysis using Machine Learning

Submitted by

Name:- Avilipsa Mahapatra

Roll No.:- **23051664**

School of Computer Engineering
KIIT - DU



KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY (KIIT)

Deemed to be University U/S 3 of UGC Act, 1956

Table of Contents

1	Introduction	3
2	Problem Statement	4
3	Python Package Used Details	5
4	Source Code.	9
5	Implementation Results.	12
6	Conclusion	14
7	References	16

Chapter 1

Introduction

The increasing levels of carbon dioxide (CO₂) emissions have become a major environmental concern worldwide. Monitoring and analyzing CO₂ emissions is essential for understanding environmental impact and making informed decisions for sustainable development.

In this project, machine learning techniques are used to analyze and predict CO₂ emissions based on various factors such as engine size, number of cylinders, and fuel consumption. Three different algorithms—Linear Regression, Decision Tree, and Random Forest—are implemented and compared.

The objective is to evaluate the performance of these algorithms and identify the most suitable model for predicting CO₂ emissions accurately.

Chapter 2

Problem Statement & Objectives

Problem Statement

CO₂ emissions depend on multiple factors and show non-linear patterns. Traditional analytical methods may fail to accurately predict such complex relationships. Therefore, there is a need to apply machine learning techniques to analyze and predict CO₂ emissions effectively.

Objectives

1. To analyze the relationship between vehicle features and CO₂ emissions
2. To implement and compare multiple machine learning algorithms
3. To evaluate model performance using metrics such as MSE, RMSE, and R² score
4. To identify the best-performing model for accurate prediction

Chapter 3

Python Packages Used Details

Name	Functions Used	Explanation
Numpy	Sqrt()	Used for numerical operations and RMSE calculation
Pandas	read_csv()	Used to load and manipulate dataset
Matplotlib	plot(), bar()	Used for data visualization
Sklearn	LinearRegression()	Implements linear regression model
Sklearn	DecisionTreeRegressor()	Implements decision tree model
Sklearn	RandomForestRegressor()	Implements random forest model
Sklearn	mean_squared_error()	Calculates prediction error
Sklearn	r2_score()	Evaluates model accuracy

Chapter 4

Source Code

NOTE:“The results and model comparisons are displayed using a webpage developed with HTML, CSS, and JavaScript. The webpage (index.html) is included in the project files.”

ML PART[.ipynb file]

PART A - ML (Fuel Dataset)

```
=====
```

```
.. code:: ipython3
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error
```

```
.. code:: ipython3
```

```
df = pd.read_csv("FuelConsumption.csv")
df.head()
```

```
.. code:: ipython3
```

```
X = df[["ENGINE_SIZE", "CYLINDERS", "FUEL_CONSUMPTION_COMB"]]
y = df["CO2_EMISSIONS"]
```

```
.. code:: ipython3
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

```
.. code:: ipython3
```

```
lr = LinearRegression()
lr.fit(X_train, y_train)
```

.. code:: ipython3

```
dt = DecisionTreeRegressor()
dt.fit(X_train, y_train)
```

.. code:: ipython3

```
rf = RandomForestRegressor()
rf.fit(X_train, y_train)
```

Compare Models using MSE

~~~~~

.. code:: ipython3

```
lr_pred = lr.predict(X_test)
dt_pred = dt.predict(X_test)
rf_pred = rf.predict(X_test)

print("Linear Regression MSE:", mean_squared_error(y_test, lr_pred))
print("Decision Tree MSE:", mean_squared_error(y_test, dt_pred))
print("Random Forest MSE:", mean_squared_error(y_test, rf_pred))
```

Error : LR > RF > DT

Graph

~~~~~

.. code:: ipython3

```
models = ["Linear", "Decision Tree", "Random Forest"]
errors = [
    mean_squared_error(y_test, lr_pred),
    mean_squared_error(y_test, dt_pred),
    mean_squared_error(y_test, rf_pred)
]

plt.bar(models, errors)
plt.title("Model Comparison")
plt.ylabel("MSE")
plt.show()
```

.. code:: ipython3

```
import pickle
pickle.dump(rf, open("model.pkl", "wb"))
```

```
.. code:: ipython3
```

```
df.describe()
```

```
.. code:: ipython3
```

```
df.corr(numeric_only=True)
```

PART B(Sector Analysis)

```
.. code:: ipython3
```

```
df2 = pd.read_csv("dataset.csv")  
df2.head()
```

```
.. code:: ipython3
```

```
df2.columns
```

```
.. code:: ipython3
```

```
import matplotlib.pyplot as plt  
  
df2.groupby("sector")["value"].sum().plot(kind='bar')  
plt.title("CO2 Emission by Sector")  
plt.show()  
plt.savefig("co2_graph.png")
```

```
.. code:: ipython3
```

```
df2.groupby("sector")["value"].sum().plot(kind='pie', autopct='%1.1f%%')  
plt.title("CO2 Contribution by Sector")  
plt.ylabel("")  
plt.show()
```

```
.. code:: ipython3
```

```
df2.groupby("date")["value"].sum().plot()  
  
plt.xticks(rotation=45)  
plt.title("CO2 Emission Over Time")  
plt.show()
```

```
.. code:: ipython3
```

```
df2["date"] = pd.to_datetime(df2["date"], dayfirst=True)  
df2["year"] = df2["date"].dt.year  
  
df2.groupby("year")["value"].sum().plot()
```



```
plt.title("CO2 Emission Over Years")
plt.xlabel("Year")
plt.ylabel("CO2 Emission")
plt.show()
```

.. code:: ipython3

```
df_india = df2[df2["country"] == "India"]

df_india.groupby("sector")["value"].sum().plot(kind='bar')
plt.title("CO2 Emission by Sector (India)")
plt.show()
```

HTML PART

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <title>CO2 Emission Analysis</title>
```

```
  <link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
  <div class="container">
```

```
    <h1>🌍 CO2 Emission Analysis using ML</h1>
```

```
    <!-- ORIGINAL GRAPH -->
```

```
    <div class="card">
```

```
      <h2>📊 Overall CO2 Emission Graph</h2>
```

```
      
```

```
    </div>
```

```
    <!-- MODEL GRAPHS -->
```

```
    <div class="card">
```

```
      <h2>🤖 Model Predictions</h2>
```

```
      <div class="graph-container">
```

```
        <div class="graph-box">
```

```
          <h3>Linear Regression</h3>
```

```
          
```

```
        </div>
```

```
        <div class="graph-box">
```

```
          <h3>Decision Tree</h3>
```

```

        
    </div>

    <div class="graph-box">
        <h3>Random Forest</h3>
        
    </div>

</div>
</div>

<div class="card">
    <h2>📊 Model Comparison (MSE)</h2>
    
</div>

<!-- COMPARISON TABLE -->
<div class="card">
    <h2>📊 Algorithm Comparison</h2>

    <table>
        <tr>
            <th>Algorithm</th>
            <th>MSE ↓</th>
            <th>RMSE ↓</th>
            <th>R2 ↑</th>
        </tr>

        <tr>
            <td>Linear Regression</td>
            <td>410.68</td>
            <td>20.27</td>
            <td>0.883</td>
        </tr>

        <tr>
            <td>Decision Tree</td>
            <td>158.64</td>
            <td>12.60</td>
            <td>0.955</td>
        </tr>

        <tr class="best">
            <td>Random Forest</td>
            <td>155.00</td>
            <td>12.45</td>
            <td>0.956</td>
        </tr>
    </table>

```

```
        <button onclick="showBest()">Find Best Model</button>
        <p id="result"></p>
    </div>
</div>
<script src="script.js"></script>

</body>

</html>
```

CSS PART

```
body {
    font-family: Arial, sans-serif;
    background-color: #f4f6f8;
    margin: 0;
    padding: 0;
}

.container {
    width: 80%;
    margin: auto;
    text-align: center;
}

h1 {
    color: #2c3e50;
    margin-top: 20px;
}

.card {
    background: white;
    padding: 20px;
    margin: 20px 0;
    border-radius: 10px;
    box-shadow: 0px 4px 10px rgba(0,0,0,0.1);
}

img {
    width: 60%;
    border-radius: 10px;
}

table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 15px;
}

table, th, td {
    border: 1px solid #ccc;
}

th {
```

```

    background-color: #3498db;
    color: white;
}

td, th {
    padding: 10px;
}

button {
    margin-top: 15px;
    padding: 10px 20px;
    background-color: #27ae60;
    color: white;
    border: none;
    border-radius: 5px;
    cursor: pointer;
}

button:hover {
    background-color: #219150;
}

.graph-container {
    display: flex;
    justify-content: space-between;
    flex-wrap: wrap;
}

.graph-box {
    width: 30%;
    text-align: center;
}

.graph-box img {
    width: 100%;
    border-radius: 10px;
}

.card img {
    width: 60%;
    border-radius: 10px;
}

.best {
    background-color: #d4edda;
    font-weight: bold;
}

```

JS PART

```

function showBest() {
    document.getElementById("result").innerHTML =
        "✅ Best Model: Random Forest(Highest Accuracy)";
}

```

Chapter 5

Implementation Results

Graph Analysis

The CO₂ emission graph shows fluctuations in emission values, indicating a non-linear relationship between input features and output.

Model Performance Comparison

Algorithm	MSE	RMSE	R ²
Linear Regression	410.68	20.27	0.883
Decision Tree	158.64	12.60	0.955
Random Forest	155.00	12.45	0.956

Model-wise Analysis

Linear Regression

The Linear Regression model produces smooth predictions but fails to capture fluctuations in the data. This results in higher error, indicating underfitting.

Decision Tree

The Decision Tree model captures non-linear patterns but produces irregular and highly fluctuating predictions, indicating overfitting.

Random Forest

The Random Forest model provides stable and accurate predictions by combining multiple trees. It shows the lowest error and highest R² score.

Final Observation

The Random Forest model outperforms the other models as it balances bias and variance effectively, making it the most suitable for this dataset.

Chapter 6

Conclusion

In this project, three machine learning algorithms were implemented to predict CO₂ emissions. The results indicate that Linear Regression is not suitable for complex datasets, while Decision Tree tends to overfit. Random Forest provides the best performance by reducing error and improving prediction accuracy.

Thus, Random Forest is identified as the most effective model for CO₂ emission prediction.

References

1. Scikit-learn Documentation
2. Pandas Documentation
3. Matplotlib Documentation
4. CO₂ Emission Dataset
 - a. <https://www.kaggle.com/datasets/saloni1712/co2-emissions>
 - b. <https://www.kaggle.com/datasets/ahmettyilmazz/fuel-consumption>